Exact values and certified lower bounds for the no-three-in-line problem in the cube

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20 August 2026 — version 1.1 (draft)
doi:10.5281/zenodo.22019279

Abstract

Let $a(n)$ be the largest number of points of the grid $\{0, 1, \ldots, n-1\}^3$ no three of which are collinear. Pór and Wood proved $a(n) = \Theta(n^2)$; no exact values appear to have been recorded, and the sequence does not appear to be in the OEIS. We determine

$$a(1), \ldots, a(6) = 1,\ 8,\ 16,\ 28,\ 40,\ 64,$$

give certified lower bounds $a(7) \geq 73$ and $a(8) \geq 93$, and prove $a(p) \geq p^2$ for every prime p by an elementary construction from an anisotropic binary quadratic form.

The method is a certified decision procedure rather than a search: a line carrying at most two lattice points forbids nothing, so admissibility is exactly "at most two chosen points on every line with three or more lattice points", which is a propositional formula on n^3 variables. Unsatisfiability is answered with a DRAT certificate that an independent checker verifies, and the lower bounds are witnesses re-verified by exhaustive integer arithmetic over every triple. We are explicit about the three separate links this requires — that the formula expresses the problem, that it is unsatisfiable, and that the symmetry pruning discards nothing — because a certificate attests only to the formula it was given, and an error in the other two links would pass through it unseen.

Along the way the optima turn out to share a layer structure that fails at exactly one of the values computed: for $n = 2, 3, 4, 6$ the two outer layers carry the planar maximum $2n$ and the inner ones $2n-2$, giving $2n^2 - 2n + 4$; at $n = 5$ two outer layers provably cannot both carry $2n$ at any total, and at $n = 7$ the corresponding profile is impossible as well. We have no explanation for this.

1 Definitions and elementary bounds

Points of $G_n := \{0, \ldots, n-1\}^3$; a set is *lawful* if no three of its points are collinear in $\mathbb{R}^3$ (equivalently, all 2×2 minors of the difference matrix vanish for no triple). Each of the $3n^2$ axis-parallel lines carries at most two points of a lawful set, and each point lies on three of them, so $|S| \leq 2n^2$; this is the trivial upper bound, attained at $n = 2$.

Proposition 1. *For every prime p, $a(p) \geq p^2$.*

Proof. Let $Q(x, y) = x^2 - dy^2$ with d a quadratic non-residue modulo p, so that $Q(a, b) \equiv 0$ only for $a \equiv b \equiv 0$ (an anisotropic binary form; one exists for every p). Put $S = \{(x, y, Q(x, y) \bmod p) : 0 \leq x, y < p\}$, of size p^2. Suppose three of its points are collinear. Their projections to the (x, y)-plane are distinct (the third coordinate is a function of the first two), hence lie on a line with primitive integer direction (a, b), and the three points are $(x_0 + t_i a,\ y_0 + t_i b,\ \cdot\,)$ with distinct integers t_i, $|t_i| < p$. Along this line the third coordinate satisfies $z \equiv Q(x_0 + ta, y_0 + tb) =$

$Q(a,b)\,t^2+\ell(t) \pmod p$ with ℓ affine, whereas collinearity in $\mathbb{R}^3$ forces z to be an affine function of t. A quadratic congruence agreeing with an affine one at three points distinct modulo p has vanishing leading coefficient, so $Q(a,b)\equiv 0$ and therefore $(a,b)=(0,0)$, a contradiction. □

Neither x^2+y^2 (anisotropic iff $p\equiv 3 \bmod 4$) nor x^2+xy+y^2 (iff $p\equiv 2 \bmod 3$) works for all p; the choice x^2-dy^2 does. The bound of Proposition 1 is not sharp: $a(5)\geq 40>25$ and $a(7)\geq 73>49$.

2 Method: a certified decision procedure

A line carrying at most two lattice points of $[n]^3$ forbids nothing, so

$$S\subseteq [n]^3 \text{ is admissible } \iff |S\cap L|\leq 2 \text{ for every line } L \text{ with } \geq 3 \text{ lattice points.} \tag{1}$$

This turns the question "is there an admissible set of size M?" into a Boolean satisfiability problem with n^3 variables — 125 for $n=5$, 216 for $n=6$ — one at-most-two cardinality constraint per rich line, and one global at-least-M constraint. The global counter is a totalizer truncated at $2n^2$; the truncation is legitimate because each of the n^2 axis-parallel lines carries at most two points, so $a(n)\leq 2n^2$. We add the 47 lexicographic leader constraints of the cube group, which preserve satisfiability because the property in (1) is invariant under that group.

We adopted this route after measuring the alternative. A prefix-split exhaustive search on 42 cores closed 5 of 204 prefixes in two hours at $n=5$, extrapolating to roughly 88 core-days; the same question is decided by a SAT solver in minutes. Speed, however, is not the reason. An exhaustive search establishes completeness by *its own* covering argument, which only its authors' code can check; an unsatisfiability answer comes with a DRAT certificate that an independent, widely used checker verifies without trusting our solver or our code at all.

Three separate links. It is worth being explicit about what is and is not certified, because the three claims below rest on different foundations.

1. *The formula expresses the problem.* Verified two independent ways: by comparing the set of triples forbidden by the rich-line constraints against the set with vanishing cross product, and by forcing each collinear triple as an assumption and requiring unsatisfiability. At $n=4$ and $n=5$ there are 376 and 1858 collinear triples; none was left unforbidden and none was forbidden spuriously. A DRAT checker cannot verify this link — it knows nothing of cubes and lines — and this is exactly where an encoding bug would hide.
2. *The formula is unsatisfiable.* Verified by `drat-trim` on the certificate emitted by `kissat`.
3. *A witness is admissible.* Verified by exhaustive integer arithmetic over all $\binom{|S|}{3}$ triples, by a program sharing no code with the search.

Links 1 and 2 give the upper bound; links 1 and 3 give the lower bound.

Calibration. Before computing anything new we reproduced every published value: $a(3)=16$ and $a(4)=28$, each with unsatisfiability at $M+1$ in under a second, independently on two machines. For instances that resist a single solver call we split into independent pieces by fixing the contents of the first columns; the split is complete because three points of a column are collinear, so subsets of size at most two exhaust it. The aggregator refuses a verdict unless every piece of the manifest is present and unsatisfiable — a missing piece and an unsatisfiable one must never look alike.

3 Exact values

n	$a(n)$	$a(n)/n^2$	how established
1	1	—	trivial
2	8	2.00	exhaustive; meets the trivial bound $2n^2$
3	16	1.78	exhaustive, and certified
4	28	1.75	exhaustive, and certified
5	40	1.60	certified
6	64	1.78	certified

The value $a(6) = 64$ appears to be new; a search of the OEIS for $1, 8, 16, 28, 40$ and for $8, 16, 28, 40, 64$ returns nothing, and the sequence itself is not in the database.

For $n \leq 4$ the values were also produced by two independent exhaustive programs whose node counts differ by a factor of about 72 ($17/3855/4.21 \cdot 10^7$ against $17/20068/3.03 \cdot 10^9$ for $n = 2, 3, 4$), so their agreement is not an artefact of a shared design.

For $n = 5$ and $n = 6$ the decisive question — does a set of 41, respectively 65, points exist? — was split into independent pieces by fixing the contents of the first two z-columns, and the heaviest pieces were split once more on the third column. Both splits are complete because three points of a column are collinear, so subsets of size at most two exhaust it. The final tally, reproduced in `logs/no3_3d/FINAL_VERDICT.txt`:

	$n = 5$, $M = 41$	$n = 6$, $M = 65$
top-level pieces	256	484
closed directly	221	450
closed through all their continuations	35	34
continuations required per piece	16	22
satisfiable pieces found	0	0
pieces left undecided	0	0

The aggregator builds the expected set of pieces from the manifest rather than from the results, treats a missing piece and an undecided one as failures, and distinguishes three outcomes rather than two: only an explicit unsatisfiable answer closes a piece, a satisfiable one anywhere would destroy the claim, and a killed or timed-out run counts as *no information* — it neither closes nor refutes. Eleven such uninformative lines occur in the $n = 6$ record, from runs interrupted when work was moved between machines; each of the pieces concerned is closed by an explicit unsatisfiable answer from another run.

The lower bounds are witnesses, checked by exhaustive integer arithmetic over all $\binom{|S|}{3}$ triples by a program sharing no code with the search: 40 points at $n = 5$ (9880 triples) and 64 points at $n = 6$ (41664 triples). For $n = 6$ two witnesses were obtained independently, by a SAT solver and by a constraint solver in decision form; they turned out to be the same set, which the stabiliser explains — it has order 24, so the configuration has only two distinct images under the 48 symmetries of the cube.

As a further check that is not part of the proof, a constraint solver was given the $n = 6$, $M = 65$ question in one piece for an hour and returned no answer. That is not evidence of impossibility — it is the solver saying it does not know — but it found no counterexample either, while at $M = 64$ it produces a solution in seconds.

4 Structure of the optima, and an anomaly at $n = 5$

Writing the layer profile of a set as the number of its points in each of the n layers perpendicular to an axis, every optimum we have found has the *same* profile along all three axes:

n	$a(n)$	layer profile (identical for all three axes)
2	8	$4, 4$
3	16	$6, 4, 6$
4	28	$8, 6, 6, 8$
5	40	$8, 8, 8, 8, 8$
6	64	$12, 10, 10, 10, 10, 12$

A layer is an $n \times n$ grid with no three collinear, so it holds at most $2n$ points. For $n = 2, 3, 4, 6$ the two outer layers attain that planar maximum and the inner ones hold $2n - 2$, giving $2 \cdot 2n + (n - 2)(2n - 2) = 2n^2 - 2n + 4$ — which reproduces $8, 16, 28$ and 64 exactly.

At $n = 5$ the pattern fails, and it fails for a structural reason rather than by accident. Prescribing the profile as a hard constraint and deciding feasibility shows that

profile at $n = 5$	verdict
$10, 8, 8, 8, 10$ (total 44, the value the pattern predicts)	infeasible
$10, 9, 8, 9, 10$; $10, 8, 9, 8, 10$; $10, 10, 10, 10, 10$; $9, 9, 9, 9, 9$	infeasible
$10, 8, 8, 8, 8$ (total 42) and $10, 7, 8, 7, 10$ (total 42)	infeasible
$\mathbf{10, 6, 8, 6, 10}$ **(total 40)**	**infeasible**
$8, 8, 8, 8, 8$ (total 40)	feasible
$n = 6$: $12, 10, 10, 10, 10, 12$ (total 64)	feasible
$n = 7$: $14, 12, 12, 12, 12, 12, 14$ (total 88)	infeasible

A caveat on how these verdicts are to be read. Prescribing the profile along all three axes at once is a restriction, so an infeasibility obtained that way would exclude only symmetric configurations, not that number of points. Every infeasibility above was therefore re-derived with the profile prescribed along a *single* axis, which excludes every configuration with that layer distribution. The satisfiable rows need no such care: a solution is a solution. The fourth line is the sharpest: at $n = 5$ two outer layers cannot both attain the planar maximum $2n$ at *any* total, so the optimum is forced onto a uniform profile. The same computation at $n = 6$ finds the analogous profile realisable, and at $n = 7$ it is again impossible. The pattern $[2n, (2n - 2)^{n-2}, 2n]$ is thus realisable at $n = 2, 3, 4, 6$ and impossible at $n = 5, 7$ — the obstruction is not an artefact of one case, and it is not an obstruction to the idea of maximal outer layers as such.

The obstruction can be priced exactly. A layer is a planar no-three-in-line configuration and so carries at most $2n$ points; requiring *both* outer layers to attain that planar maximum, and maximising the total, gives

n	both outer layers	maximum total	$a(n)$
4	8	28	28 — costs nothing
5	10	**39**	**40 — costs exactly one point**
6	12	64	64 — costs nothing

So the anomaly is not that a fitted formula occasionally fails. It is that two planar-optimal configurations placed in the outer planes are compatible with the spatial optimum at $n = 4$ and $n = 6$ and incompatible at $n = 5$, and incompatible by a single point.

We do not have a proof of why. A count of the planar configurations attaining $2n$ points suggests rigidity: the 4×4 grid admits 11 such configurations in 4 classes under the square's symmetries, the 5×5 grid admits 32 in only 5 classes, and the 6×6 grid admits 50 in 11 classes. We offer this as an observation accompanied by its data, not as an argument. One natural mechanism can be ruled out: the midpoint of a segment joining the two outer layers is a lattice point when n is odd, and must stay empty, but an exact count shows the middle layer still retains up to 16 free cells — above its own planar ceiling of 10 — so that constraint does not bind.

5 Certified lower bounds for $n = 7, 8$

n	$a(n) \geq$	$/n^2$	source of the witness
7	73	1.49	invariant under a cyclic axis permutation (order 3)
8	93	1.45	invariant under the full group of axis permutations (order 6)

Each configuration is re-verified by testing all $\binom{|S|}{3}$ triples with integer cross products. These are lower bounds only; we make no claim about how far they are from the truth, and we expect them to be weak, since a randomised search is a poor instrument at this size. The $n = 7$ bound illustrates the point: it stood at 71 from randomised search and rose to 73 within the hour once the search was restricted to configurations invariant under a cyclic permutation of the axes. Restricting to a subgroup collapses the cells into orbits and shrinks the search by the order of the group; it is legitimate for lower bounds only, since a configuration found is a configuration, while failure to find one among the symmetric ones says nothing about the rest. The size of the group has an optimum in the middle rather than at either end, and the optimum is a balance: the group must be large enough to make the problem solvable and small enough that good configurations still fall inside the symmetric stratum. At $n = 8$:

order of the group	best found
2	82, 84
3	88
4	88, 90
6	**92, 92, 93**
8	80, 88
12	88
24	88 (proved optimal within its stratum)
48	80 (proved optimal within its stratum)

At $|G| = 48$ only 20 orbits remain, the solver settles the stratum in seconds, and the answer is 80, because genuinely good configurations do not have that much symmetry. Note also that the order alone does not decide it: two different groups of order 8 gave 80 and 88. We record this as data. An earlier draft of this note stated instead that a larger group simply gives a better result — a pattern read off the first four rows, which the remaining four contradict. (An earlier draft of this note quoted 72 and 90 here. Those numbers came from a run whose witness was never saved, and when we came to attach witnesses to them the journals we could reproduce gave 71 and 89. We report the values we can exhibit.) Two indications. First, the layer structure of the smaller optima suggests $2n^2 - 2n + 4 = 88$ at $n = 7$ — and although we have shown that the corresponding *profile* is impossible, that leaves the value itself untouched. Second, at $n = 6$ the same randomised search reached 64, which the certified computation then confirmed to be optimal, so the instrument is not hopeless either; we simply do not know which case $n = 7$ resembles.

For the record we also measured what the abandoned route would have cost, though the measurement deserves a caveat. Knuth's random-probe estimator, calibrated against the exact tree at $n = 4$, put an unaided exhaustive proof at $n = 6$ at some $4.6 \cdot 10^{22}$ nodes, and that figure is what sent us to a decision procedure. We later found the plain estimator unreliable on trees of this family — on a sibling problem the mean of twenty thousand probes was 89% a single probe, with the median five orders of magnitude below the mean — and replaced it by a stratified version that agrees with the exact tree at $n = 4$ to within 0.1%. The $4.6 \cdot 10^{22}$ was produced by the plain estimator and has not been recomputed with the stratified one, so it should be read as an order of magnitude and nothing finer. It says nothing about the difficulty of the problem in any case, only about the difficulty of one way of attacking it.

6 Reproducibility, and what is *not* claimed

All programs, journals, witnesses, manifests and the verification protocol are public: `https://github.com/iwasborninbali/saturation`. The protocol `docs/VERIFICATION.md` is written for a reader who trusts none of our code and gives the commands in order.

The following are the limits of what is established here, stated so that a reader need not infer them.

- The upper bounds for $n = 5, 6$ rest on one SAT solver. The instances that a single call decides carry DRAT certificates verified by `drat-trim`, which removes the dependence on the solver; the instances that had to be split do not carry stored certificates, because the certificates run to tens of gigabytes. They are regenerated deterministically on demand, and the case split, its completeness and its aggregation are each verified separately.
- We have no proof of the upper bounds independent of the propositional encoding. Our own exhaustive enumerator does not reach $n = 5$: it closed 24 of 4096 prefixes at 15–33 billion nodes each, with the full traversal on the order of 10^{14} nodes. It corroborates the lower bound only.
- The verification of the symmetry pruning shows that our *implementation* discards nothing; it is not a proof of the underlying lex-leader argument, which is standard but which we did not reprove.
- The layer-structure anomaly at $n = 5$ and $n = 7$ is a computation, not an explanation. A count of the planar configurations attaining $2n$ points suggests rigidity (11 configurations in 4 classes for 4×4; 32 in only 5 classes for 5×5; 50 in 11 classes for 6×6), and one natural mechanism — the lattice midpoint between two outer layers — is ruled out by an exact count. Beyond that we do not know.

AI/tool-use disclosure

The searches, the programs and the text were produced by autonomous AI agents (Anthropic Claude) working under the direction of the author of record, who posed the question, chose what to compute, decided what to claim and takes responsibility for the content. Two agents worked with deliberately different program designs and audited each other; several errors reported in the repository were caught that way and not by their author, including one — an unverified symmetry pruning — that no certificate, witness check or coverage check would have detected.

The work is arranged so that this should not matter. Every claim above is meant to be checkable by a third party without trusting the agents or the author: certificates by `drat-trim`, witnesses by brute-force integer arithmetic, coverage by a mechanical aggregator that refuses a verdict on a missing or undecided piece, and the encoding by comparison against the direct collinearity criterion. Where a claim is not checkable in that way, it is listed in the preceding section.